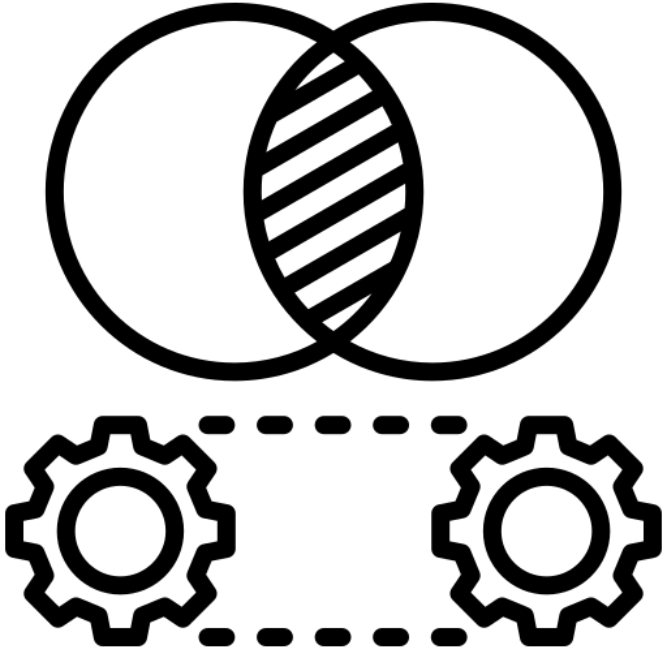
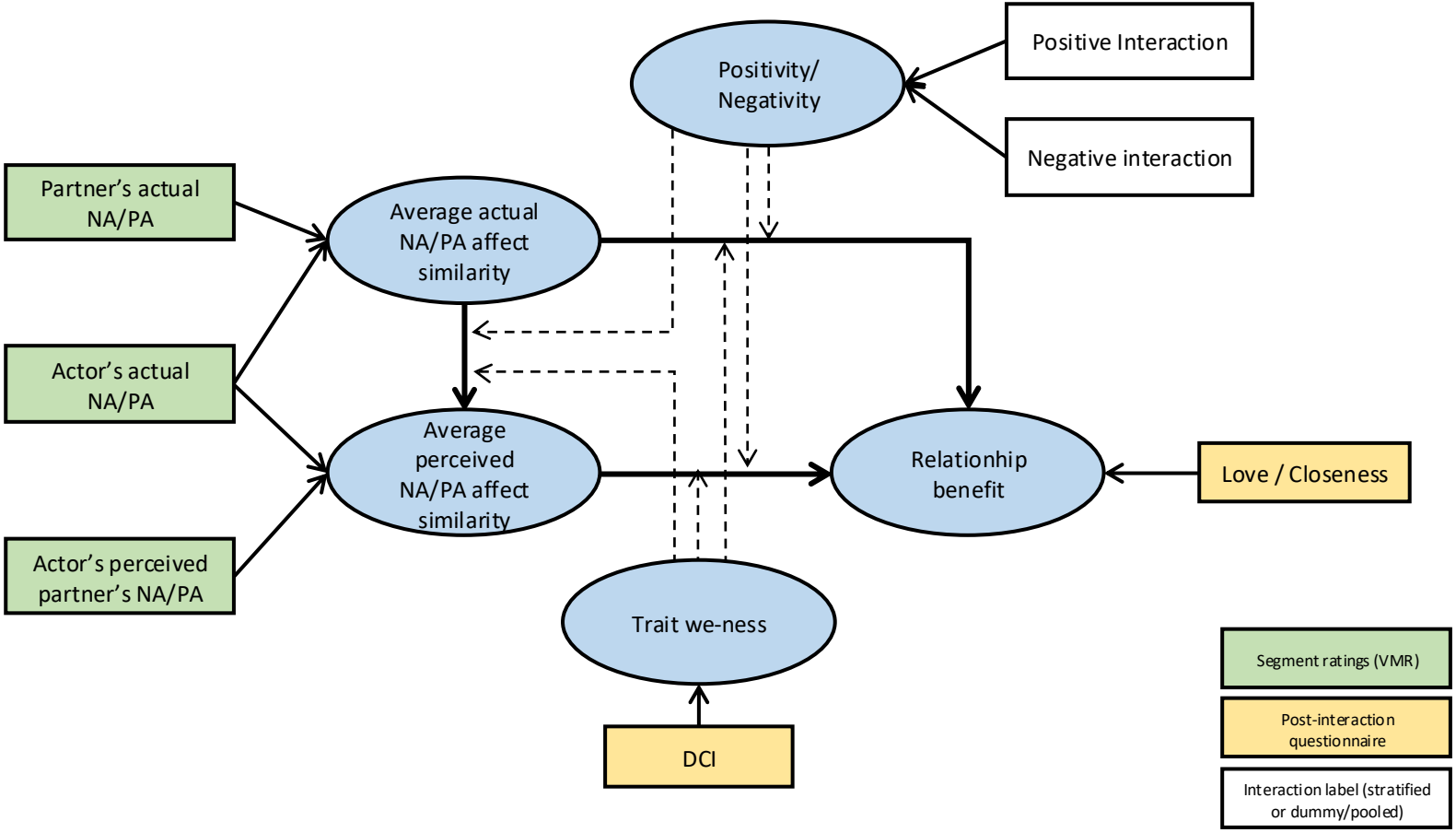
Feeling the Same, Seeing the Same

Affect similarity, perception, and relationship evaluations



Results from study component 2 (VMR and Background Questionnaire data)

Feeling versus seeing – Study Component 2



Overview of preregistered analyses

For both the positive and negative laboratory interactions:

1a
- Does higher interaction-averaged actual PA similarity relate to higher post-interaction love/closeness?
- Does higher interaction-averaged actual NA similarity relate to lower post-interaction love/closeness?

H1.1-H1.2: Yes

1b
Does adding interaction-averaged perceived similarity to the model, add something above and beyond and 'steal away' from the actual similarity-love/closeness association?

H1.3.PA – H1.3.NA: Yes

1c
Do interaction-averaged actual and perceived similarity interact?

Exploratory

2
- Does interaction-averaged perceived similarity mediate the association?
- Is there a remaining direct link between interaction-averaged actual similarity and love/closeness

H2.1-H2.2: Yes, at least partially

Exploratory

3a
Does the link between interaction-averaged actual similarity and post-interaction love/closeness depend on trait we-ness?

H3.1-H3.2: Yes, amplify (PA) or attenuate (NA)

3b
Does the link between interaction-averaged actual similarity and post-interaction love/closeness depend on interaction valence (positive/negative)

H3.3-H3.4: Mismatch hypothesis

1 Base analyses (H1)

1a

- H1.1 Average Actual PA similarity → Love
- H1.1 Average Actual PA similarity → Closeness
- H1.2 Average Actual NA similarity → Love
- H1.2 Average Actual NA similarity → Closeness

1b

- H1.3.Average Actual + Perceived PA similarity → Love
- H1.3.Average Actual + Perceived PA similarity → Closeness
- H1.3.Average Actual + Perceived NA similarity → Love
- H1.3.Average Actual + Perceived NA similarity → Closeness

1c

- H1.3.PA with interaction term*
- H1.3.NA with interaction term*

Preregistered hypotheses
Exploratory (Preregistered)

H1.1 – Actual PA Similarity → Post-interaction Love

Models h11_PA_love_pos, h11_PA_love_neg

H1.1: Higher interaction-averaged actual PA similarity (APAS) between partners is associated with higher post-interaction love (O).

$O = b_0 + \beta_1 \cdot APAS + \varepsilon$, [APAS = Actual PA Similarity, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

Positive Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	85.97	3.24	[79.59, 92.35]	< .001
APAS (β_1)	0.459	0.168	[0.127, 0.791]	.007

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
109.6	176.3	285.9	.385

Negative Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	89.05	4.64	[79.88, 98.23]	< .001
APAS (β_1)	0.379	0.161	[0.060, 0.697]	.020

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
78.2	336.9	415.0	.189

Key Conclusions

- H1.1 supported for love in both contexts
- The effect is slightly larger in positive context (.459 vs. .379)
- High intercepts (85–89) indicate strong baseline love even after negative interaction
- Partners diverge more after negative interaction (ICC = .189) compared to the positive interaction (ICC = .385).
- On the other hand, there is less difference between dyads after negative interaction (between dyad variance 78.2) compared to the positive interaction (between dyad variance 109.6)
- The total variance increases from positive (285.9) to negative (415.0) interaction context, meaning that there is more heterogeneity in post-interaction love evaluations. This possibly suggests that negative interactions are more destabilizing.

H1.1 – Actual PA Similarity → Post-interaction Closeness

Models h11_PA_close_pos, h11_PA_close_neg

H1.1: Higher interaction-averaged actual PA similarity (APAS) between partners is associated with higher post-interaction closeness (O).

$O = b_0 + \beta_1 \cdot APAS + \varepsilon$, [APAS = Actual PA Similarity, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

Positive Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	89.77	3.07	[83.73, 95.81]	< .001
APAS (β_1)	0.376	0.154	[0.071, 0.681]	.016

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
87.8	154.3	242.1	.363

Negative Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	92.07	4.88	[82.43, 101.70]	< .001
APAS (β_1)	0.432	0.175	[0.085, 0.779]	.016

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
162.6	257.8	420.4	.387

Key Conclusions

- H1.1 supported for closeness in both contexts
- The effect is slightly larger in negative context (.432 vs. .376)
- High intercepts (89–92) indicate strong baseline closeness, even the highest after negative interaction
- The total variance increases from positive (242.1) to negative (420.4) interaction context, meaning that there is more heterogeneity in post-interaction closeness evaluations. This possibly suggests that negative interactions are more destabilizing.
- Contrary to the models where love is the outcome variable, both between dyad as within dyad (residual) variance increases from positive to negative interaction.

H1.2 – Actual NA Similarity → Post-interaction Love

Models h12_NA_love_pos, h12_NA_love_neg

H1.1: Higher interaction-averaged actual NA similarity (ANAS) between partners is associated with lower post-interaction love (O). ✗

$O = b_0 + \beta_1 \cdot ANAS + \varepsilon$, [ANAS = Actual NA Similarity, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

Positive Interaction ✗

Fixed Effects				
Parameter	**Est.**	**SE**	**95% CI**	**p**
Intercept (β_0)	**91.58**	**3.24**	**[85.21, 97.95]**	**< .001**
ANAS (β_1)	**0.352**	**0.133**	**[0.089, 0.615]**	**.009**

Variance decomposition			
Between dyad	**Residual** (within dyad)	**Total**	**Partner correlation (ICC)**
110.5	176.3	286.9	.385

Negative Interaction ✗

Fixed Effects				
Parameter	**Est.**	**SE**	**95% CI**	**p**
Intercept (β_0)	**77.53**	**2.52**	**[72.57, 82.49]**	**< .001**
ANAS (β_1)	-0.141	0.211	[-0.559, 0.276]	.504

Variance decomposition			
Between dyad	**Residual** (within dyad)	**Total**	**Partner correlation (ICC)**
91.3	336.9	428.1	.213

Key Conclusions

- H1.2 not supported for love in both the positive and the negative interaction:
  - In the negative interaction, the slope is non-significant
  - In the positive interaction, the slope is positive instead of negative, suggesting attunement rather than strain
- High intercepts (77–91) indicate strong baseline love, lowest after negative interaction
- The total variance increases substantially from positive to negative contexts, meaning more heterogeneity in post-interaction love and suggesting higher destabilization.
- The variance patterns replicate those from H1.1 – Love.
  - The between dyad variance decreases from positive (110.5) to negative interaction (91.3), meaning that dyads are more alike
  - Meanwhile, the within variance/residual increases from positive (176.3) to negative interaction (336.9)
  - Partners agree more on love after the positive interaction

H1.2 – Actual NA Similarity → Post-interaction Closeness

Models h12_NA_close_pos, h12_NA_close_neg

H1.1: Higher interaction-averaged actual NA similarity (ANAS) between partners is associated with lower post-interaction closeness (O). ✗

$O = b_0 + \beta_1 \cdot ANAS + \varepsilon$, [ANAS = Actual NA Similarity, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

Positive Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**93.91**	**2.89**	**[88.23, 99.59]**	**< .001**
ANAS (β_1)	**0.383**	**0.119**	**[0.148, 0.617]**	**.002**

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
80.9	154.3	235.2	.344

Negative Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**78.97**	**2.75**	**[73.56, 84.38]**	**< .001**
ANAS (β_1)	-0.157	0.231	[-0.612, 0.299]	.497

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
179.8	257.8	437.6	.411

Key Conclusions

- H1.2 for closeness replicates the H1.2 love pattern.
- Hypotheses not supported in both interactions:
  - In the negative interaction, the slope is non-significant
  - In the positive interaction, the slope is positive instead of negative, suggesting attunement rather than strain
- High intercepts (79–94) indicate strong baseline closeness, lowest after negative interaction
- The total variance increases substantially from positive to negative contexts, meaning more heterogeneity in post-interaction closeness and suggesting higher destabilization.
- The variance patterns replicate those from H1.1 – Closeness.
  - The between dyad variance increases from positive (80.9) to negative interaction (179.8), indicating higher dyad differences.
  - The within variance/residual increases from positive (154.3) to negative interaction (257.8)
  - Partners agree more on love after the positive interaction

H1.1 / H1.2 Summary

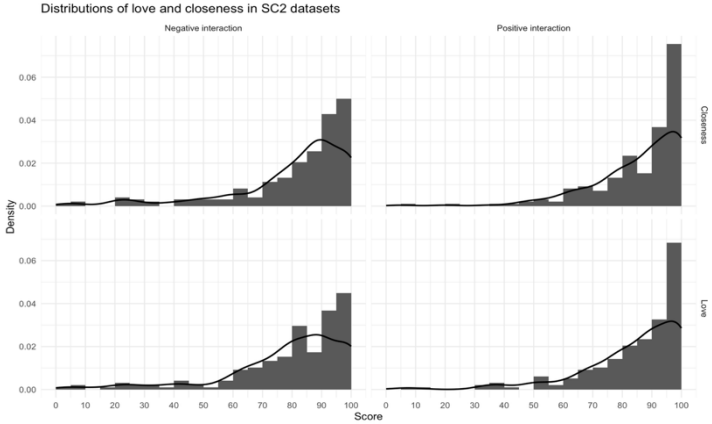
- **H1.1 is supported across both outcomes and both interaction contexts.**
  - Actual PA similarity during the interaction consistently predicts higher post-interaction love and closeness.
  - The association does not depend on whether the interaction was a negative or a positive.
- **H1.2 is not supported. The hypothesized negative association is absent entirely.**
  - In the positive interaction, the association is significant but in the positive direction: sharing more negative affect during a positive discussion predicts higher love and closeness, not lower.
  - In the negative interaction, Actual NA similarity is uninformative for both outcomes (both $p > .49$).
- **For both H1.1 and H1.2, the negative interactions show a higher total variance for the negative interactions compared to the positive interactions, suggesting a more broadly destabilization.**
- **For love as outcome, there is more agreement between partners after the positive interaction compared to the negative interaction (higher ICCs, couples are more internally consistent)**

Model diagnostics (H1.1)

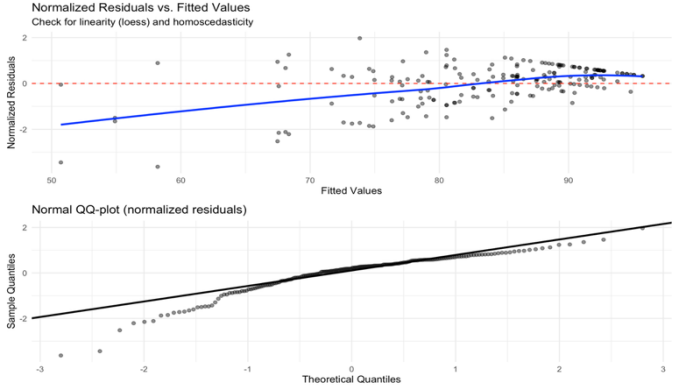
- **Diagnostic checks were performed for H1.1**
  - **Residuals vs. Fitted:** Slight negative fan shape at high fitted values, consistent with ceiling compression on the 0–100 scale. The LOESS smoother is below zero for lower fitted values (50-75) because of a few isolated observations with very low fitted values (around 50–60) and large negative residuals.
  - **QQ-plot:** Left-tail departure (heavy left tail), consistent with the ceiling effect: a few individuals report much lower post-interaction evaluations. Right tail is truncated at the ceiling.
- **Overall:**
  None of the model diagnostics suggest severe enough violations to make conclusions by default unreliable but the ceiling-driven violations warrant caution for inference.

- The patterns are structural consequences of the bounded outcome scale combined with high baseline love and closeness in the sample, so diagnostics are not repeated for subsequent models.

Distribution of love and closeness in the SC2 dataset



Example: diagnostics for the h11_PA_love_neg model



H1.3 – Actual + Perceived PA Similarity → Post-interaction Love

Models h13_PA_love_pos, h13_PA_love_neg

H1.3: When averaged perceived PA similarity is added, it should predict post-interaction love (O) in the same positive direction as averaged actual PA similarity in H1.1, and the actual PA similarity effect should attenuate ✗

$O = b_0 + \beta_1 \cdot APAS + \beta_2 \cdot PPAS + \varepsilon$, [APAS = Actual PA Similarity, PPAS = Perceived PA Similarity, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

Positive Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**85.52**	**1.76**	**[82.03, 89.01]**	**< .001**
APAS (β_1)	0.386	0.205	[-0.022, 0.793]	.063
PPAS (β2)	0.105	0.170	[-0.232, 0.441]	.538

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
107.6	178.4	286.0	.376

Negative Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**89.10**	**4.63**	**[79.90, 98.29]**	**< .001**
APAS (β_1)	**0.444**	**0.221**	**[0.005, 0.883]**	**.048**
PPAS (β2)	-0.081	0.190	[-0.459, 0.296]	.670

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
78.3	338.5	416.7	.188

Key Conclusions

- H1.3 for PA Similarity is not supported for love in either interaction context
- In the positive interaction, the coefficients for both predictors become non-significant
- In the negative interaction, the effect of actual similarity remains significant (and somewhat larger than in step 1a where it is 0.379). The effect of perceived similarity is not significant.
- Variance decompositions and ICCs virtually unchanged from H1.1 (step 1a)

H1.3 – Actual + Perceived PA Similarity → Post-interaction Closeness

Models h13_PA_close_pos, h13_PA_close_neg

H1.3: When averaged perceived PA similarity is added, it should predict post-interaction closeness (O) in the same positive direction as averaged PA actual similarity in H1.1, and the actual PA similarity effect should attenuate ✗

$O = b_0 + \beta_1 \cdot APAS + \beta_2 \cdot PPAS + \varepsilon$, [APAS = Actual PA Similarity, PPAS = Perceived PA Similarity, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

Positive Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**86.28**	**1.61**	**[83.08, 89.47]**	**< .001**
APAS (β_1)	0.215	0.188	[-0.158, 0.588]	.256
PPAS (β2)	0.229	0.156	[-0.082, 0.540]	.147

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
87.4	153.3	240,7	.363

Negative Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**92.08**	**5.04**	**[82.09, 102.08]**	**< .001**
APAS (β_1)	**0.456**	**0.224**	**[0.011, 0.901]**	**.045**
PPAS (β2)	-0.031	0.176	[-0.379, 0.318]	.863

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
162.9	259.4	422.3	.386

Key Conclusions

- H1.3 for PA Similarity is not supported for closeness in either interaction context
- In the positive interaction, the coefficients for both predictors become non-significant
- In the negative interaction, the effect of actual similarity remains significant (at a similar level than in step 1a where it is 0.432). The effect of perceived similarity is not significant.
- Variance decompositions and ICC virtually unchanged from H1.1 (step 1a)

H1.3 – Actual + Perceived NA Similarity → Post-interaction Love

Models h13_NA_love_pos, h13_NA_love_neg

H1.3: When averaged perceived NA similarity is added, it should predict love (O) in the same positive direction as averaged actual NA similarity in H1.1, and the actual NA similarity effect should attenuate ✗

$O = b_0 + \beta_1 \cdot ANAS + \beta_2 \cdot PNAS + \varepsilon$, [ANAS = Actual NA Similarity, PNAS = Perceived NA Similarity, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

Positive Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**91.61**	**3.21**	**[85.23, 98.00]**	**< .001**
ANAS (β_1)	0.332	0.176	[-0.017, 0.681]	.062
PNAS (β2)	0.025	0.141	[-0.256, 0.3005]	.862

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
110.0	177.9	287.9	.382

Negative Interaction ✗

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**77.31**	**2.55**	**[79.90, 98.29]**	**< .001**
ANAS (β_1)	-0.217	0.256	[-0.726, 0.292]	.400
PNAS (β2)	0.108	0.207	[-0.302, 0.518]	.603

Variance decomposition			
Between dyad	Residual (within dyad)	Total	Partner correlation (ICC)
95.5	335.1	430.6	.222

Key Conclusions

- H1.3 for NA Similarity is not supported for love in either interaction context
- For both interactions, the coefficients for both predictors become non-significant
- Variance decompositions and ICC virtually unchanged from H1.2 (step 1a)

H1.3 – Actual + Perceived NA Similarity → Post-interaction Closeness

Models h13_NA_close_pos, h13_NA_close_neg

H1.3: When averaged perceived NA similarity is added, it should predict closeness (O) in the same positive direction as averaged actual NA similarity in H1.1, and the actual NA similarity effect should attenuate ✗

$$O = b_0 + \beta_1 \cdot ANAS + \beta_2 \cdot PNAS + \varepsilon, \quad [ANAS = \text{Actual NA Similarity}, PNAS = \text{Perceived NA Similarity}, \text{random intercept at dyad level } b_0 = \beta_0 + v_{0,d}]$$

Positive Interaction ✗

Fixed Effects				
Parameter	**Est.**	**SE**	**95% CI**	**p**
Intercept (β_0)	**93.68**	**2.88**	**[87.96, 99.41]**	**< .001**
ANAS (β_1)	**0.528**	**0.159**	**[0.212, 0.843]**	**.001**
PNAS (β2)	-0.177	0.156	[-0.434, 0.080]	.174

Variance decomposition			
Between dyad	**Residual** (within dyad)	**Total**	**Partner correlation (ICC)**
84.3	151.3	235.5	.358

Negative Interaction ✗

Fixed Effects				
Parameter	**Est.**	**SE**	**95% CI**	**p**
Intercept (β_0)	**78.92**	**2.76**	**[73.44, 84.40]**	**< .001**
ANAS (β_1)	-0.175	0.267	[-0.705, 0.355]	.513
PNAS (β2)	0.026	0.193	[-0.357, 0.410]	.892

Variance decomposition			
Between dyad	**Residual** (within dyad)	**Total**	**Partner correlation (ICC)**
180.6	259.2	439.7	.411

Key Conclusions

- H1.3 for NA Similarity is not supported for closeness in either interaction context
- In the negative interaction, the coefficients for both predictors become non-significant
- In the positive interaction, the effect of actual similarity remains significant (and somewhat larger than in step 1a where it is 0.383). The effect of perceived similarity is not significant.
- Variance decompositions and ICC virtually unchanged from H1.1 (step 1a)

H1.3 Summary

- **H1.3 is not supported across both outcomes and both interaction contexts**
- **Perceived similarity is non-significant in all models**
- **When adding perceived similarity to the models, actual similarity loses significance in some models**
- Actual and perceived similarity overlap structurally, because actor affect contributes to both indices:
  - Their interaction-averages are moderately to strongly correlated (r = .62 - .75)
  - When both predictors are entered simultaneously, this leads to inflated standard errors (VIF = 1.62 - 2.31) and reduced estimation precision.
  - A previously significant effect can therefore become nonsignificant
- This overlap/collinearity also exists in SC1, but is less consequential there because:
  - SC1 has many repeated ESM beeps per person, yielding a much larger effective sample size; standard errors are smaller to begin with, so the additional inflation from collinearity is proportionally less impactful.
  - Aggregating across VMR segments to a single value per interaction compresses variation and increases the predictor correlation, because segment-level noise that might differentiate the two indices gets averaged out.

Exploratory – Actual x Perceived PA Similarity → Post-interaction Love

Models im_PA_love_pos, im_PA_love_neg

Exploratory: Adding the interaction Actual x Perceived PA Similarity to models h13_PA_love_pos and h13_PA_love_neg and test whether the association of each form of similarity with post-interaction love (O) is conditional on the level of the other.

$$O = b_0 + \beta_1 \cdot APAS + \beta_2 \cdot PPAS + \beta_3 \cdot APAS \times PPAS + \varepsilon \qquad [\text{random intercept at dyad level } b_0 = \beta_0 + v_{0,d}]$$

Positive Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**83.77**	**1.57**	**[80.65, 86.89]**	**< .001**
APAS (β_1)	0.393	0.207	[-0.019, 0.804]	.061
PPAS (β2)	0.099	0.171	[-0.240, 0.438]	.564
APAS x PPAS (β_3)	0.003	0.013	[-0.023, 0.029]	.798

Negative Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**78.57**	**1.88**	**[74.84, 82.29]**	**< .001**
APAS (β_1)	0.437	0.223	[-0.006, 0.881]	.053
PPAS (β2)	-0.067	0.200	[-0.463, 0.329]	.738
APAS x PPAS (β_3)	0.003	0.013	[-0.022, 0.028]	.808

Key Conclusions

- For both interaction contexts, the interaction term is non-significant

Exploratory – Actual x Perceived PA Similarity → Post-interaction Closeness

Models im_PA_close_pos, im_PA_close_neg

Exploratory: Adding the interaction Actual x Perceived PA Similarity to models h13_PA_close_pos and h13_PA_close_neg and test whether the association of each form of similarity with post-interaction closeness (O) is conditional on the level of the other.

$$O = b_0 + \beta_1 \cdot APAS + \beta_2 \cdot PPAS + \beta_3 \cdot APAS \times PPAS + \varepsilon \qquad [\text{random intercept at dyad level } b_0 = \beta_0 + v_{0,d}]$$

Positive Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**85.44**	**1.44**	**[82.60, 88.30]**	**< .001**
APAS (β_1)	0.221	0.190	[-0.157, 0.598]	.249
PPAS (β2)	0.225	0.158	[-0.088, 0.438]	.537
APAS x PPAS (β_3)	0.003	0.012	[-0.021, 0.027]	.796

Negative Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**80.67**	**2.00**	**[76.70, 84.64]**	**< .001**
APAS (β_1)	**0.464**	**0.226**	**[0.015, 0.914]**	**.043**
PPAS (β2)	-0.048	0.186	[-0.416, 0.321]	.798
APAS x PPAS (β_3)	-0.004	0.013	[-0.029, 0.022]	.774

Key Conclusions

- For both interaction contexts, the interaction term is non-significant
- For the negative interaction, APAS remains significant (p = .043) and at the same level as in h13_PA_close_neg and h11_PA_close_neg

Exploratory – Actual x Perceived NA Similarity → Post-interaction Love

Models im_NA_love_pos, im_NA_love_neg

Exploratory: Adding the interaction Actual x Perceived NA Similarity to models h13_NA_love_pos and h13_NA_love_neg and test whether the association of each form of similarity with post-interaction love (O) is conditional on the level of the other.

$$O = b_0 + \beta_1 \cdot APAS + \beta_2 \cdot PPAS + \beta_3 \cdot APAS \times PPAS + \varepsilon \qquad [\text{random intercept at dyad level } b_0 = \beta_0 + v_{0,d}]$$

Positive Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**83.86**	**1.71**	**[80.46, 87.26]**	**< .001**
APAS (β_1)	0.333	0.176	[-0.018, 0.683]	.063
PPAS (β2)	0.026	0.142	[-0.256, 0.308]	.856
APAS x PPAS (β_3)	0.001	0.010	[-0.019, 0.021]	.928

Negative Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**79.12**	**1.79**	**[75.56, 82.68]**	**< .001**
APAS (β_1)	-0.200	0.258	[-0.712, 0.312]	.441
PPAS (β2)	-0.096	0.208	[-0.317, 0.509]	.645
APAS x PPAS (β_3)	-0.008	0.018	[-0.043, 0.028]	.673

Key Conclusions

- For both interaction contexts, the interaction term is non-significant

Exploratory – Actual x Perceived NA Similarity → Post-interaction Closeness

Models im_NA_close_pos, im_NA_close_neg

Exploratory: Adding the interaction Actual x Perceived NA Similarity to models h13_NA_close_pos and h13_NA_close_neg and test whether the association of each form of similarity with post-interaction closeness (O) is conditional on the level of the other.

$$O = b_0 + \beta_1 \cdot ANAS + \beta_2 \cdot PNAS + \beta_3 \cdot ANAS \times PNAS + \varepsilon \qquad [\text{random intercept at dyad level } b_0 = \beta_0 + v_{0,d}]$$

Positive Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**85.09**	**1.54**	**[82.04, 88.14]**	**< .001**
ANAS (β_1)	**0.531**	**0.159**	**[0.215 0.847]**	**.001**
PNAS (β2)	-0.169	0.130	[-0.423, 0.090]	.198
ANAS x PNAS (β_3)	0.005	0.009	[-0.013, 0.024]	.548

Negative Interaction

Fixed Effects				
Parameter	Est.	SE	95% CI	p
Intercept (β_0)	**80.46**	**1.93**	**[76.62, 84.29]**	**< .001**
ANAS (β_1)	-0.171	0.269	[-0.706, 0.364]	.527
PNAS (β2)	0.024	0.195	[-0.363, 0.410]	.903
ANAS x PNAS (β_3)	-0.002	0.018	[-0.037, 0.034]	.921

Key Conclusions

- For both interaction contexts, the interaction term is non-significant
- For the positive interaction, ANAS remains significant (p = .001) and at the same level as in h13_NA_close_pos and h11_NA_close_pos

Exploratory – Actual x Perceives Similarity Summary

- **All eight interaction terms are statistically zero, based on the sample**
- **Main effect patterns largely replicate steps 1a and 1b**
- **Two predictors remain significant throughout steps 1a, 1b, and 1c:**
    - Actual NA similarity predicting closeness in the positive interaction
    - Actual PA similarity predicting closeness in the negative interaction

Summary: Steps 1a, 1b, and 1c

			Positive Interaction			Negative Interaction		
Outcome	**Affect Similarity**	**Predictor**	**1a**	**1b**	**1c**	**1a**	**1b**	**1c**
Closeness	**PA Similarity**	**Actual**	**0.376***	0.215	0.221	**0.432***	**0.456***	**0.464***
		Perceived		0.229	0.224		-0.031	-0.048
		Actual × Perceived			0.003			-0.004
	NA Similarity	**Actual**	**0.383****	**0.528****	**0.531****	-0.156	-0.175	-0.171
		Perceived		-0.177	-0.169		0.026	0.024
		Actual × Perceived			0.005			-0.002
Love	**PA Similarity**	**Actual**	**0.459****	0.386	0.393	**0.379***	**0.444***	0.437
		Perceived		0.105	0.099		-0.081	-0.067
		Actual × Perceived			0.003			0.003
	NA Similarity	**Actual**	**0.352****	0.332	0.333	-0.141	-0.217	-0.200
		Perceived		0.025	0.026		0.108	0.096
		Actual × Perceived			0.001			-0.008

Note. Coefficients with *$p < .05$, **$p < .01$, ***$p < .001$. Significant coefficients ($p < .05$) in **bold**. Step 1a includes actual similarity only. Step 1b adds perceived similarity. Step 1c adds the actual × perceived interaction (predictors grand-mean centered within context). All models include a dyad-level random intercept. Empty cells indicate that the predictor is not in the model at that step.

2 Mediation analyses (H2)

- H2.1 Mediation of PA similarity → Love
- H2.1 Mediation of PA similarity → Closeness
- H2.2 Mediation of NA similarity → Love
- H2.2 Mediation of NA similarity → Closeness

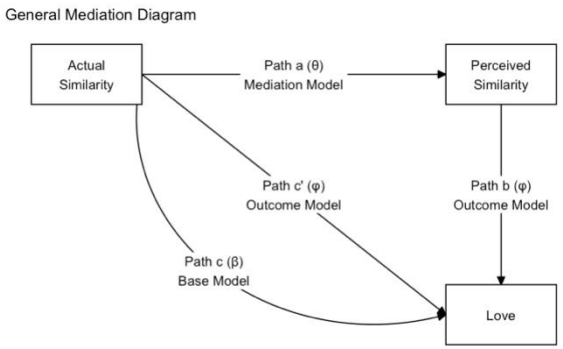
H2.1 – PA similarity → Love mediation

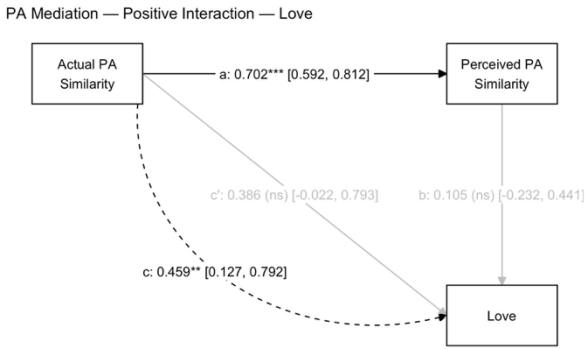
Models h11_PA_love_pos, h11_PA_love_neg, h21_PA_med_pos, h21_PA_med_neg, h21_PA_out_love_pos, and h21_PA_out_love_neg

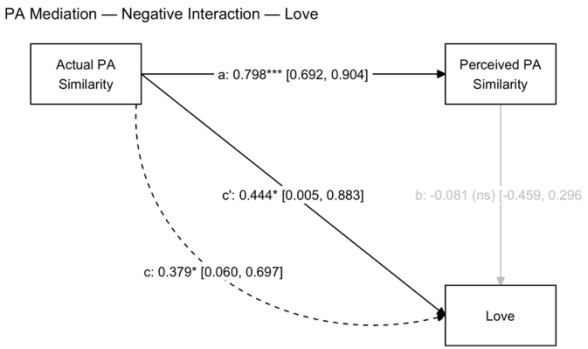
H2.1: The association between interaction-averaged actual PA similarity and post-interaction love (O) is at least partially mediated by average perceived PA similarity. ❌

Mediator model: $PPAS = t_0 + \theta_1 . APAS + \varepsilon$, [random intercept]

Outcome model: $O = f_0 + \varphi_1 \cdot PPAS + \varphi_2 . APAS + \varepsilon$ [random intercept]







Key Conclusions

- **H2.1 is not supported for PA similarity → love in either interaction**
- The a-paths are strong
- The b-path don't reach significance
- The indirect and direct effects are as previously established

Monte Carlo indirect effects: PA similarity to Love mediation

Interaction valence	Indirect effect	95% MC CI	p-value
Positive	0.074	[-0.161, 0.313]	0.544
Negative	-0.065	[-0.364, 0.235]	0.677

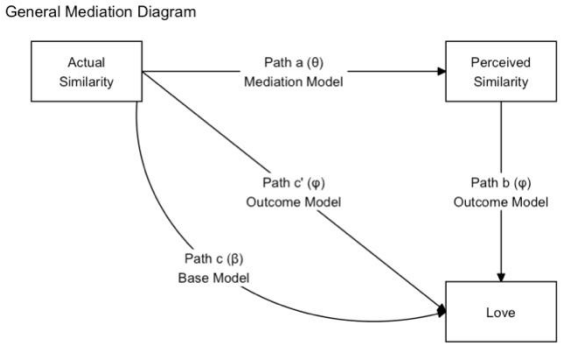
Note. CIs based on 20,000 Monte Carlo draws.

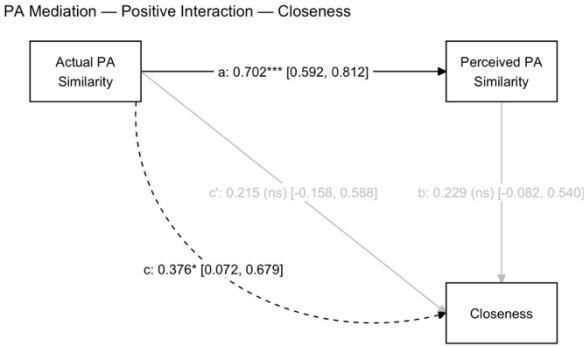
H2.1 – PA similarity → Closeness mediation

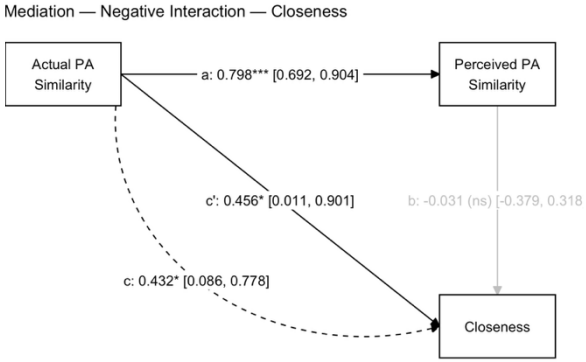
Models h11_PA_close_pos, h11_PA_close_neg, h21_PA_med_pos ,h21_PA_med_neg, h21_PA_out_close_pos, and h21_PA_out_close_neg

H2.1: The association between averaged actual PA similarity and post-interaction closeness (O) is at least partially mediated by average perceived PA similarity. ✗

Mediator model: $PPAS = t_0 + \theta_1 . APAS + \varepsilon$, [random intercept]
Outcome model: $O = f_0 + \varphi_1 \cdot PPAS + \varphi_2 . APAS + \varepsilon$ [random intercept]







Key Conclusions

- **H2.1 is not supported for PA similarity → closeness in either interaction**
- The a-paths are strong
- The b-path don't reach significance
- The indirect and direct effects are as previously established

Monte Carlo indirect effects: PA similarity to Closeness mediation

Interaction valence	Indirect effect	95% MC CI	p-value
Positive	0.161	[-0.049, 0.435]	0.151
Negative	-0.024	[-0.263, 0.284]	0.858

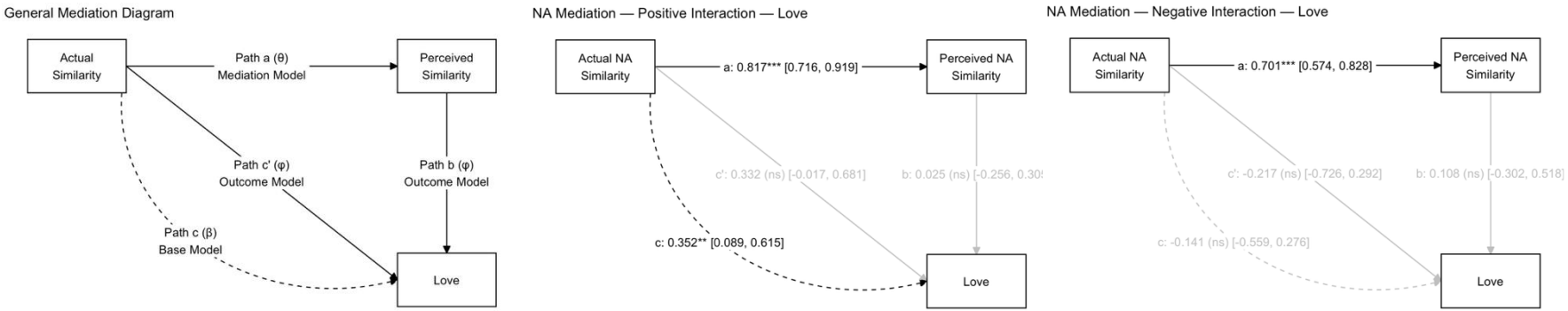
Note. CIs based on 20,000 Monte Carlo draws.

H2.2 – NA similarity → Love mediation

Models h11_NA_love_pos, h11_NA_love_neg, h22_NA_med_pos, h22_NA_med_neg, h22_NA_out_love_pos, and h22_NA_out_love_neg

H2.1: The association between interaction-averaged actual NA similarity and post-interaction love (O) is at least partially mediated by average perceived NA similarity. ❌

Mediator model: $PNAS = t_0 + \theta_1 . ANAS + \varepsilon$, [random intercept]
Outcome model: $O = f_0 + \varphi_1 \cdot PNAS + \varphi_2 . ANAS + \varepsilon$ [random intercept]



Key Conclusions

- **H2.2 is not supported for NA similarity → love in either interaction**
- The a-paths are strong
- The b-path don't reach significance
- The indirect and direct effects are as previously established

Monte Carlo indirect effects: NA similarity to Love mediation

Interaction valence	Indirect effect	95% MC CI	p-value
Positive	0.020	[-0.207, 0.249]	0.856
Negative	0.076	[-0.209, 0.368]	0.605

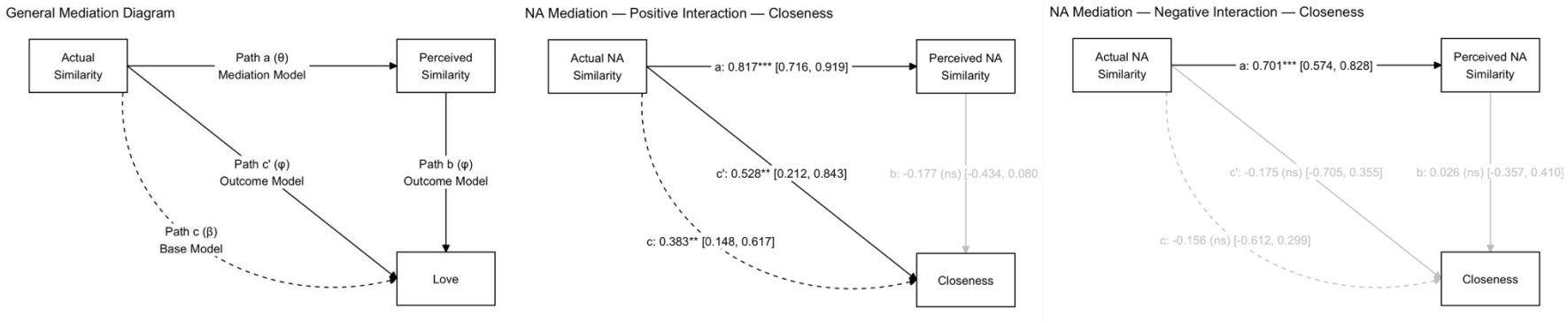
Note. CIs based on 20,000 Monte Carlo draws.

H2.2 – NA similarity → Closeness mediation

Models h11_NA_close_pos, h11_NA_close_neg, h22_NA_med_pos, h22_NA_med_neg, h22_NA_out_close_pos, and h22_NA_out_close_neg

H2.1: The association between averaged actual NA similarity and post-interaction closeness (O) is at least partially mediated by average perceived NA similarity. ✗

Mediator model: $PNAS = t_0 + \theta_1 . ANAS + \varepsilon$, [random intercept]
Outcome model: $O = f_0 + \varphi_1 \cdot PNAS + \varphi_2 . ANAS + \varepsilon$ [random intercept]



Key Conclusions

- **H2.2 is not supported for NA similarity → closeness in either interaction**
- The a-paths are strong
- The b-path don't reach significance
- The indirect and direct effects are as previously established

Monte Carlo indirect effects: NA similarity to Closeness mediation

Interaction valence	Indirect effect	95% MC CI	p-value
Positive	-0.145	[-0.311, 0.070]	0.173
Negative	0.019	[-0.205, 0.336]	0.891

Note. CIs based on 20,000 Monte Carlo draws.

H2.1 / H2.2 Mediation Summary

		PA Similarity (H2.1)		NA Similarity (H2.2)	
Outcome	**Path**	**Negative Interaction**	**Positive Interaction**	**Negative Interaction**	**Positive Interaction**
Love/Closeness	**a-path**	**0.798*****	**0.702*****	**0.701*****	**0.817*****
Love	**b-path**	-0.081	0.105	0.108	0.025
	c-path (indirect)	-0.065	0.074	0.076	0.020
	c'-path (direct)	**0.444***	0.386	-0.217	0.332
Closeness	**b-path**	-0.031	0.229	0.026	-0.177
	c-path (indirect)	-0.024	0.161	0.019	-0.145
	c'-path (direct)	**0.456***	0.215	-0.175	**0.528****

Note. Coefficients with * $p < .05$, ** $p < .01$, *** $p < .001$. Significant coefficients ($p < .05$) in **bold**. a-path: actual similarity predicting perceived similarity (shared across outcomes). b-path: perceived similarity predicting outcome controlling for actual similarity. c-path (indirect): product of a-path and b-path, tested via Monte Carlo simulation (20,000 draws). c'-path (direct): direct effect of actual similarity controlling for perceived similarity. All models include a dyad-level random intercept.

- For all mediation models, the a-path is always strong: actual PA similarity is perceived accurately.
- The b-path (from perceived similarity to the outcome) never reaches significance. Perceived similarity never predicts the outcome once actual similarity is controlled for.
- There is thus also never an indirect (c) significant path
- The direct paths (c') follow previosuy established patterns

3a Moderation by trait we-ness (H3.1 + H3.2)

- H3.1 Actual PA similarity × We-ness → Love
- H3.1 Actual PA similarity × We-ness → Closeness
- H3.1 Perceived PA similarity × We-ness → Love
- H3.1 Perceived PA similarity × We-ness → Closeness

- H3.2 Actual NA similarity × We-ness → Love
- H3.2 Actual NA similarity × We-ness → Closeness
- H3.2 Perceived NA similarity × We-ness → Love
- H3.2 Perceived NA similarity × We-ness → Closeness

H3.1 / H3.2 Summary

H3.1: Higher trait we-ness amplifies the positive association between averaged PA similarity (actual and perceived) and post-interaction love and closeness.
H3.2: Higher trait we-ness attenuates the negative association between averaged NA similarity (actual and perceived) and post-interaction love and closeness.

				Interaction Valence	
Outcome	**Affect**	**Similarity**	**Parameter**	**Positive Interaction**	**Negative Interaction**
Closeness	PA Similarity	Actual	Similarity	0.196	**0.385***
			We-ness	**10.932*****	-0.173
			Similarity × We-ness	-0.447	-0.441
		Perceived	Similarity	**0.293***	0.159
			We-ness	**13.736*****	12.898
			Similarity × We-ness	-0.180	0.019
	NA Similarity	Actual	Similarity	**0.265***	-0.255
			We-ness	9.679	**15.205***
			Similarity × We-ness	-0.129	0.191
		Perceived	Similarity	0.045	0.009
			We-ness	**13.309***	**18.267*****
			Similarity × We-ness	-0.020	**1.284****
Love	PA Similarity	Actual	Similarity	0.300	**0.341***
			We-ness	**8.301***	-0.657
			Similarity × We-ness	-0.476	-0.383
		Perceived	Similarity	**0.273***	0.144
			We-ness	**11.828*****	12.334
			Similarity × We-ness	-0.234	0.089
	NA Similarity	Actual	Similarity	0.251	-0.212
			We-ness	10.701	**14.204***
			Similarity × We-ness	0.022	0.307
		Perceived	Similarity	0.140	0.005
			We-ness	10.546	**14.128****
			Similarity × We-ness	-0.018	0.758

Note. Coefficients with * p < .05, ** p < .01, *** p < .001. Significant coefficients (p < .05) in **bold**. Similarity: main effect of similarity at average we-ness. We-ness: main effect of trait we-ness at average similarity. Sim × We-ness: interaction term (H3.1 for PA, H3.2 for NA; expected positive). All models include a dyad-level random intercept.

Key Conclusions

- We-ness direct effects
  - In the **PA similarity models**, couples higher in trait we-ness report more love/closeness after **positive interactions**. In the negative interactions, there is no significance for these models.
  - In the **NA similarity models**, couples higher in trait we-ness report more love/closeness after **negative interactions**. This suggests that shared negativity is functioning as mutual engagement ('we are in this together')
- There is only 1 significant moderation, for the perceived NA similarity to closeness association.

3b Moderation by interaction valence (positive/negative) (H3.3 + H3.4)

- Stratified models (1a models for positive/negative data subset)

- H3.3 Actual PA similarity × Interaction valence → Love
- H3.3 Actual PA similarity × Interaction valence → Closeness
- H3.3 Perceived PA similarity × Interaction valence → Love
- H3.3 Perceived PA similarity × Interaction valence → Closeness

- H3.4 Actual NA similarity × Interaction valence → Love
- H3.4 Actual NA similarity × Interaction valence → Closeness
- H3.4 Perceived NA similarity × Interaction valence → Love
- H3.4 Perceived NA similarity × Interaction valence → Closeness

Preregistered hypotheses
Exploratory (Preregistered)

Stratified Models - Descriptive Comparison

H3.3: The positive association between averaged PA similarity (actual and perceived) and post-interaction love and closeness is stronger (more positive) in the negative interaction than in the positive interaction. ✗

H3.4: The negative association between interaction-level NA similarity (actual and perceived) and post-interaction love and closeness is stronger (more negative) in the positive interaction than in the negative interaction. ✗

			Interaction Valence	
Outcome	**Affect**	**Similarity**	**Positive**	**Negative**
Love	**PA Similarity**	**Actual**	**0.459****	**0.379***
		Perceived	**0.292***	0.181
	NA Similarity	**Actual**	**0.352****	-0.141
		Perceived	0.201	0.005
Closeness	**PA Similarity**	**Actual**	**0.376***	**0.432***
		Perceived	**0.334***	0.195
	NA Similarity	**Actual**	**0.383****	-0.156
		Perceived	0.109	-0.040

Note. Coefficients with * $p < .05$, ** $p < .01$, *** $p < .001$. Significant coefficients ($p < .05$) in **bold**. Actual similarity coefficients are from Step 1a models. Perceived similarity coefficients are from analogous bivariate models substituting perceived for actual similarity. All models include a dyad-level random intercept and are estimated separately within each interaction context.

- Hypotheses: mismatch effect. PA similarity will be most beneficial during negative interactions (H3.3). NA similarity will be most detrimental during positive interactions (H3.4).
- H3.3: only for closeness and actual PA similarity
- H 3.4: there is no negative association to begin with.
- The formal test (next slides) determines whether these descriptive context differences are statistically reliable.

H3.3 / H3.4 Summary

H3.3: The positive association between averaged PA similarity (actual and perceived) and post-interaction love and closeness is stronger (more positive) in the negative interaction than in the positive interaction. ✗

H3.4: The negative association between interaction-level NA similarity (actual and perceived) and post-interaction love and closeness is stronger (more negative) in the positive interaction than in the negative interaction. ✗

Outcome	Affect	Similarity	Parameter	Estimate
Closeness	PA Similarity	Actual	Similarity	**0.311****
			Interaction Valence	-2.102
			Similarity × Interaction Valence	-0.063
		Perceived	Similarity	**0.284****
			Interaction Valence	-1.030
			Similarity × Interaction Valence	-0.075
	NA Similarity	Actual	Similarity	**-0.318***
			Interaction Valence	**7.986****
			Similarity × Interaction Valence	0.313
		Perceived	Similarity	-0.164
			Interaction Valence	3.521
			Similarity × Interaction Valence	0.038
Love	PA Similarity	Actual	Similarity	**0.226***
			Interaction Valence	0.839
			Similarity × Interaction Valence	0.184
		Perceived	Similarity	0.140
			Interaction Valence	1.720
			Similarity × Interaction Valence	0.236
	NA Similarity	Actual	Similarity	**-0.424****
			Interaction Valence	**11.623*****
			Similarity × Interaction Valence	**0.547****
		Perceived	Similarity	-0.158
			Interaction Valence	**7.580*****
			Similarity × Interaction Valence	0.251

Note. Coefficients with * $p < .05$, ** $p < .01$, *** $p < .001$. Significant coefficients ($p < .05$) in **bold**. Similarity: slope for the negative interaction (context = 0). Context: difference in intercept between positive and negative interaction. Sim × Context: difference in similarity slope between contexts (H3.3 for PA, H3.4 for NA; expected negative). All models include dyad-level and person-level random intercepts.

Key Conclusions

- **Interaction valence is coded as a dummy variable C (0 for the negative interaction; 1 for the positive interaction):**

$$O = b_0 + \beta_1 \cdot S + \beta_2 \cdot C + \beta_3 \cdot S \times C + \varepsilon$$

[S = Similarity, C = Interaction Context/Valence, random intercept at dyad level $b_0 = \beta_0 + v_{0,d}$]

- Therefore:
  - The coefficient for 'similarity' (β_1) is the slope for similarity in the negative interaction
  - $\beta_1 + \beta_3$ is the slope for similarity in the positive interaction
- β_3, the coefficient of the interaction term is therefore hypothesized to be negative for all models (it is what is added to the slope when going from the negative interaction to the positive)
- Only one interaction term is found to be significant, but with the opposite sign of what was hypothesized